Amna Jamal

amnajkhattak@gmail.com | Website | GitHub
OpenReview | LinkedIn | Pakistan

Research Interests

Mechanistic Interpretability for AI Safety, with a focus on diffusion models and language models.

Education

COMSATS University Islamabad Islamabad, Pakistan
Bachelor of Science in Computer Science Feb. 2022 – Feb. 2026

Experience

Algoverse Jul. 2025 – Present
AI Researcher

- Led mechanistic interpretability research on internal representations in Diffusion Transformers using causal ablations, activation interventions, probing, attention analysis, and MLP analysis, leading to the first-author paper *Diffusion Transformers Use Sink Registers*.
- Extending this work to study sparse internal structures in Diffusion Transformers and their implications for model efficiency.
- Led a second research direction on concept suppression in Diffusion Transformers using steering vectors.

PRISM AI Safety Fellowship Jun. 2026 – Present
Research Fellow — Safe-by-Design AI

- Work under Jobst Heitzig on whether LLMs can construct executable formal world models for safety-by-design AI agents.
- Develop and evaluate stochastic-game and partially observable stochastic-game representations generated by LLMs.

PRISM AI Safety Fellowship Jun. 2026 – Oct. 2026
Program Support

- Support two research teams studying conflicting information in LLMs and AI preference drift during training.
- Manage the teams and contribute to technical and non-technical blockers arising during research execution.

BlueDot Impact May 2026 – Jun. 2026
Technical AI Safety Researcher

- Investigated whether linear refusal probes capture genuine model representations or superficial prompt features.

ResearchCircle Sep. 2025 – Present
Founder, Campus & AI Safety Lead

- Founded and lead a 700+ member undergraduate research community spanning computer science, AI, mathematics, physics, and related fields.
- Lead the AI Safety sub-circle, organizing paper-reading sessions, technical discussions, and workshops to increase engagement with AI safety research in Pakistan.

Webrange Solutions Sep. 2024 – Oct. 2025
AI Intern

- Built a generative-AI platform that lets fashion businesses automate ad creation, with built-in content-moderation tools, project and campaign management for running their advertising.

Research Papers

Diffusion Transformers Use Sink Registers
First author
XAI4Science Workshop at AAAI 2026 [Paper]

One Carrier Isn't Enough: Redundancy-Aware Concept Suppression in Diffusion Transformers
First author
AI4Good Workshop at Neurips 2026 [Paper]

Evaluating LLM-Generated World Models for Neuro-Symbolic Reasoning
Co-author
NEmo Workshop at Neurips 2026 [Paper]

DUC-Bench: A Guideline-Grounded Benchmark for Evaluating Evidence-Responsive Decision Updating in Medical Large Language Models
Co-author
AI4Good and AISciK Workshops at Neurips 2026 [Paper]

Position: Contradiction Resolution in LLMs Is a Type Error, Not a Knowledge Problem
Co-author
Under review [Paper]

Academic Service

Peer Reviewer, NeurIPS 2026 Workshops 2026
Trustworthy AI for Good (AI4GOOD) Workshop
World Models in Physical AI (WM-PAI) Workshop

Technical Skills

Mechanistic Interpretability: activation extraction and hooks, causal ablation, activation patching, activation transplantation, steering vectors, linear probing, attention/QK analysis, residual-stream analysis, representation analysis

Machine Learning: PyTorch, Hugging Face Transformers, Hugging Face Diffusers, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib

Programming & Research Tools: Python, Git/GitHub, Linux, LaTeX/Overleaf, MATLAB, Qiskit, Java, JavaScript, SQL, HTML/CSS, React, Node.js, Express.js, MongoDB, MySQL

Training & Academic Activities

IAIFI Summer School 2026 2026
NSF Institute for Artificial Intelligence and Fundamental Interactions

Oxford Machine Learning School (OxML 2026) 2026
MLx Representation Learning & Generative AI

ML Summer School 2026
Cohere Labs

BlueDot Impact Programs 2026
Technical AI Safety, AGI Strategy, and Frontier AI Governance

IBM Qiskit Global Summer School 2025
Achieved Quantum Excellence

Global Physics Summit 2025
Participant

5th Riazuddin Memorial School on Quantum Computing 2025

Peer Tutor 2024
Differential Equations and Linear Algebra

Harvard Summer School 2019
Particle Physics